Restaurant Recommendation System with Geospatial Analysis

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Abstract

This project builds an AI-based restaurant recommendation system on the Zomato restaurants dataset. It combines **content-based filtering** (TF-IDF vectors of cuisines compared with cosine similarity) with **geospatial analysis** (Haversine distance and DBSCAN density clustering) to recommend restaurants that are similar in food type and close to the user. On 9,045 cleaned restaurants, the hybrid model reached a Precision@5 of 0.94 against 0.34 for random recommendations, while recommending places 2.2 km away on average (random: 11.1 km). DBSCAN found 172 restaurant hotspots.

1. Problem Statement

People in a new city face thousands of restaurants and little personalized guidance. Existing lists rank places by popularity only and ignore where the user is. The goal is a recommender that answers: *(a) which restaurants are similar to the one I like?* and *(b) which of them are near me?*

2. Dataset

Zomato Restaurants dataset (Kaggle): 9,551 restaurants with 21 columns. The project uses ten of them: Restaurant ID, Restaurant Name, Country Code, City, Address, Locality, Locality Verbose, Longitude, Latitude and Cuisines. After cleaning (removing rows with missing cuisines or coordinates, invalid (0,0) coordinates and duplicates), **9,045 restaurants** remain.

Column	Type	Use in the project
Restaurant ID / Name	ID / text	Identification and display
Country Code, City, Locality	categorical	Filtering by place; hotspot naming
Longitude, Latitude	numeric	Haversine distance, DBSCAN clustering
Cuisines	text (multi-label)	TF-IDF features for similarity

3. Methodology

3.1 Preprocessing. Coordinates are converted to numbers, invalid and duplicate rows are removed, text is trimmed.

3.2 Content-based filtering. Each restaurant's cuisine list becomes a TF-IDF vector, so rare cuisines weigh more than common ones. Similarity between two restaurants is the cosine of their vectors.

3.3 Geospatial proximity. The great-circle distance *d* (km) is computed with the Haversine formula and converted to a proximity score *exp(-d / 5)*.

3.4 Hybrid score. *Score = 0.7 x cosine_similarity + 0.3 x proximity*, ranking candidates from the same city. A second function recommends a wanted cuisine around the user's GPS point within a chosen radius.

3.5 Hotspot detection. DBSCAN with the Haversine metric (eps = 0.5 km, min_samples = 5) groups restaurants into dense food districts without choosing the number of clusters.

3.6 Evaluation. For 497 random query restaurants, the top-5 recommendations from the same city are checked: a recommendation is relevant if it shares at least one cuisine with the query. Hybrid, content-only and random strategies are compared.

4. Results

4.1 Recommendation quality

Strategy	Precision@5	Avg. distance of recommendations (km)
Hybrid (cuisine + distance)	0.944	2.22
Content-only (cuisine)	0.947	9.54
Random baseline	0.342	11.09

Mean Jaccard overlap of the hybrid recommendations: 0.792. The hybrid model keeps almost all of the cuisine accuracy of the content-only model (0.944 vs 0.947) while recommending places about 4 times closer (2.2 km vs 9.5 km), and is far better than random.

4.2 Geospatial hotspots

DBSCAN found **172** dense restaurant clusters; 1,582 restaurants are isolated. The five largest:

Hotspot area	City	Restaurants	Most common cuisine
Defence Colony	New Delhi	940	North Indian
Sector 18	Noida	487	North Indian
Rajouri Garden	New Delhi	474	North Indian
Laxmi Nagar	New Delhi	318	North Indian
Mukherjee Nagar	New Delhi	256	North Indian

Clusters are named after their most common locality; DBSCAN can join neighbouring dense areas into one cluster.

4.3 Example recommendations

Because you like **Days of the Raj** (New Delhi):

Recommended restaurant	Area	Cuisines	Sim.	Km	Score
Essex Village Garden	Essex Farms	North Indian, Chinese, Continental, Seafood	0.944	1.018	0.905
Tribe	Vasant Kunj	Continental, Seafood, North Indian	1.000	3.521	0.848
Typsy Crow	SDA	North Indian, Continental	0.627	1.256	0.672

Because you like **K Lab** (Gurgaon):

Recommended restaurant	Area	Cuisines	Sim.	Km	Score
Tea Halt	Sohna Road	Cafe	0.608	0.701	0.686
CHA	Sector 31	Beverages, Fast Food	0.698	2.388	0.675
Cafe Coffee Day	Ninex City Mart Mall, Sohna Road	Cafe	0.608	0.926	0.675

Because you like **Ali Baba Caves** (Noida):

Recommended restaurant	Area	Cuisines	Sim.	Km	Score
Tea Lounge - Jaypee Greens	Jaypee Greens Golf & Spa Resort, Surajpur	Cafe	0.848	1.578	0.813
Cafe Coffee Day	Greater Noida	Cafe	0.848	2.076	0.792
Serendipity Cafe	Greater Noida	Cafe	0.848	2.090	0.791

5. Implementation

Python 3 with pandas and NumPy (data), scikit-learn (TF-IDF, cosine similarity, DBSCAN), Matplotlib (charts) and ReportLab (report). The class *RestaurantRecommender* exposes *recommend_similar(name, city)* and *recommend_nearby(lat, lon, cuisine, radius_km)*. The notebook was run on Kaggle and the code is on GitHub (setota-123).

6. Conclusion and Future Work

Combining text features (cuisines) with geographic information gives recommendations that are relevant and nearby, without needing user history (no cold-start problem). Future work: add ratings, price range and votes (present in the dataset) to the score; collaborative filtering when user data exists; a web or mobile app with a live map; evaluation with real user feedback.

7. References

Zomato Restaurants Data, Kaggle. | Pedregosa et al., Scikit-learn: Machine Learning in Python, JMLR 2011. | Ester et al., DBSCAN, KDD 1996. | Sinnott, Virtues of the Haversine, Sky and Telescope 1984.